

40.1 Marginal versus conditional models

A clinician asks what a treatment would do to *this* patient's odds of recovery; a health minister asks what it would do to the recovery rate of the *whole population*. In a linear model the two questions have the same answer, because averaging a straight line over patients returns the same straight line. As soon as the link is nonlinear they do not, and no amount of data removes the difference: it is a property of the model, not of the sample. This section sets up the two targets and proves that the population-averaged coefficient of a logistic model is strictly smaller in magnitude than the individual one.

40.1.1 Two coefficients, one model

Throughout this chapter the data are **clustered**: units $i = 1, \dots, m$ each carry n_i observations $Y_{i1}, \dots, Y_{in_i}$, with $n = \sum_i n_i$ — a patient observed repeatedly, a school with its pupils, a firm across years. The simplest model that makes observations within a cluster dependent gives each cluster its own level U_i , with the link g of [Chapter 34](#):

$$g\{E(Y_{ij} | U_i)\} = \alpha + \beta x_{ij} + U_i, \quad U_i \sim N(0, \tau^2), \quad (40.1.1)$$

with $Y_{i1}, \dots, Y_{in_i}$ conditionally independent given U_i . Here β is a **conditional**, or *subject-specific*, coefficient: it compares two covariate values *within one cluster*, with that cluster's level held wherever it happens to be. Write it β^C when the distinction matters.

The other target concerns the response averaged over clusters. Write $h = g^{-1}$ and

$$m(x) = E\{h(\alpha + \beta x + U)\}, \quad U \sim N(0, \tau^2), \quad (40.1.2)$$

for the **marginal mean function**: the mean response at x in a cluster drawn at random. The **marginal**, or *population-averaged*, coefficients are the ones a model with the same link would report if fitted to the population: the solution (α^M, β^M) of

$$E \left[\begin{pmatrix} 1 \\ X \end{pmatrix} \{m(X) - h(\alpha^M + \beta^M X)\} \right] = \mathbf{0}, \quad (40.1.3)$$

the expectation being over the covariate distribution, and assumed to have a unique solution — which it does whenever $E\{(1, X)^\top(1, X)\}$ is nonsingular and the marginal mean function stays away from the ends of the range of h . This is the population version of the likelihood equations Eq. 34.3.1 with the dispersion and the prior weights dropped; for a canonical link the remaining factor $h'(\eta)/V(\mu)$ is identically one, so for the logit, log and identity links of this section (α^M, β^M) is exactly what an ordinary generalized linear model fitted to the clustered data estimates consistently — a point [Section 40.4](#) returns to. For a binary covariate [Eq. 40.1.3](#) is solved by matching the two means exactly, $h(\alpha^M) = m(0)$ and $h(\alpha^M + \beta^M) = m(1)$, so that β^M is the log odds ratio between the two marginal probabilities.

Warning. Neither coefficient is “the” right one and neither is an approximation to the other. They are answers to different questions, and a paper that reports one while discussing the other has made a substantive error, not a numerical one.

40.1.2 The attenuation inequality

Proposition 40.1.1. Marginal and conditional coefficients

Let U have mean zero and let h be the inverse link of the model [Eq. 40.1.1](#).

- (a) **Identity link.** If $h(\eta) = \eta$ then $m(x) = \alpha + \beta x$, so $\alpha^M = \alpha$ and $\beta^M = \beta$ for every distribution of U with mean zero. The two targets coincide.
- (b) **Log link.** If $h(\eta) = e^\eta$ and $E e^U < \infty$, then $m(x) = E(e^U) e^{\alpha + \beta x}$, so $\beta^M = \beta$ and $\alpha^M = \alpha + \log E e^U$. For $U \sim N(0, \tau^2)$ the shift is $\tau^2/2$. Only the intercept moves.
- (c) **Probit link.** If $h = \Phi$ and $U \sim N(0, \tau^2)$ then

$$m(x) = \Phi\left(\frac{\alpha + \beta x}{\sqrt{1 + \tau^2}}\right), \quad (40.1.4)$$

so the marginal mean function is again a probit, with $\alpha^M = \alpha/\sqrt{1 + \tau^2}$ and $\beta^M = \beta/\sqrt{1 + \tau^2}$.

- (d) **Logit link.** Let $h(\eta) = e^\eta/(1 + e^\eta)$, let U be any random variable with $0 < \text{Var}\{h(\theta + U)\} < \infty$ for every θ , and put $P_\theta = h(\theta + U)$ and $F(\theta) = g\{E P_\theta\}$, the marginal linear predictor. Then

$$F'(\theta) = 1 - \frac{\text{Var}(P_\theta)}{E(P_\theta) E(1 - P_\theta)} \in (0, 1). \quad (40.1.5)$$

Consequently, if $\beta \neq 0$, then for any $x_1 < x_2$ the marginal log odds ratio is strictly attenuated:

$$0 < \text{sign}(\beta) \{g m(x_2) - g m(x_1)\} < |\beta| (x_2 - x_1).$$

In particular a binary covariate has $0 < |\beta^M| < |\beta^C|$.

Proof.

- (a) $E(\alpha + \beta x + U) = \alpha + \beta x$, and this is exactly [Eq. 40.1.3](#) with h the identity.
- (b) $E e^{\alpha + \beta x + U} = e^{\alpha + \beta x} E e^U$. For a normal U , Theorem 3.2.1 gives $E e^U = e^{\tau^2/2}$. The right-hand side is again of the form $\exp(\alpha^M + \beta^M x)$, so it solves [Eq. 40.1.3](#) exactly.
- (c) Let $Z \sim N(0, 1)$ be independent of U and write $\eta = \alpha + \beta x$. Then

$$E \Phi(\eta + U) = E \Pr(Z \leq \eta + U \mid U) = \Pr(Z - U \leq \eta),$$

and $Z - U \sim N(0, 1 + \tau^2)$, which is [Eq. 40.1.4](#). Again the result is of the assumed form, so it solves [Eq. 40.1.3](#).

- (d) Write $P_\theta = h(\theta + U)$ and $\bar{p}(\theta) = E P_\theta$. The logistic inverse link satisfies $h' = h(1 - h)$, so differentiating under the expectation (legitimate because $0 \leq h' \leq 1/4$, so the difference quotients are dominated) gives $\bar{p}'(\theta) = E\{P_\theta(1 - P_\theta)\}$. Since $F(\theta) = \log \bar{p}(\theta) - \log\{1 - \bar{p}(\theta)\}$,

$$F'(\theta) = \frac{\bar{p}'(\theta)}{\bar{p}(\theta)\{1 - \bar{p}(\theta)\}} = \frac{E(P_\theta) - E(P_\theta^2)}{E(P_\theta) E(1 - P_\theta)}.$$

The denominator is $E(P_\theta) - \{E(P_\theta)\}^2$, so numerator minus denominator equals $\{E(P_\theta)\}^2 - E(P_\theta^2) = -\text{Var}(P_\theta)$, which gives [Eq. 40.1.5](#). The ratio is positive because $P_\theta \in (0, 1)$ makes $E\{P_\theta(1 - P_\theta)\} > 0$, and it is strictly less than one because $\text{Var}(P_\theta) > 0$. Integrating F' from $\alpha + \beta x_1$ to $\alpha + \beta x_2$, an interval of positive length

because $\beta \neq 0$, gives the stated inequality; for a binary covariate the left-hand side is β^M by the remark after [Eq. 40.1.3](#). ■

Part (d) needs no normality of U , no particular covariate and no asymptotics; it is an identity about the logistic function. Its factor $F'(\theta)$ is one minus the ratio of the variance of the individual probabilities to the variance a single Bernoulli variable with the average probability would have, so the more the clusters differ, the flatter the population curve.

Idea. A nonlinear link and a random effect do not commute. Averaging $h(\eta + U)$ over U gives a *different function* of η , and for the logit and probit it is a flatter one. The linear and log links are the exceptions, because for them averaging changes at most an additive or multiplicative constant.

40.1.3 How large is the difference?

The probit factor in [Eq. 40.1.4](#) is exact; the logit has no closed form, but the two links are so nearly proportional that a good approximation comes from pretending it does. Matching a standard logistic distribution to a normal one scaled by $c = 16\sqrt{3}/(15\pi) = 0.5881$ and applying [Eq. 40.1.4](#) gives the classical factor

$$\beta^M \approx \frac{\beta^C}{\sqrt{1 + c^2\tau^2}}, \quad c = \frac{16\sqrt{3}}{15\pi}, \quad (40.1.6)$$

due to Zeger, Liang and Albert (1988). It is an approximation, not a theorem.

Example. Attenuation in a logistic random-intercept model

Take $\alpha = 0$, $\beta^C = 1$ and a covariate $X \sim N(0, 1)$, and solve [Eq. 40.1.3](#) numerically. With $\tau = 0.5$ the marginal slope is 0.9502; with $\tau = 1$ it is 0.8379; with $\tau = 2$ it is 0.6142. The approximation [Eq. 40.1.6](#) gives 0.8620 and 0.6477 in the last two cases, too large by about three and five per cent. The probit factor at $\tau = 1$ is 0.7071, far smaller — but that is a different model, with a different meaning for β^C . The covariate distribution matters little: at $\tau = 1$ a binary covariate gives a marginal log odds ratio of 0.8318. [Figure 40.1.1\(a\)](#) shows why the approximation works at all — the average of the conditional curves is almost exactly a flatter logistic curve — and panel (b) tracks β^M/β^C as the clusters become more heterogeneous.

```
import numpy as np
from numpy.polynomial.hermite_e import hermegauss
from scipy.optimize import root
from scipy.stats import norm

def expit(z):
    return 1.0 / (1.0 + np.exp(-z))

# nodes and weights that integrate g(U) over U ~ N(0, 1): sum_k w_k g(z_k)
z_node, z_w = hermegauss(80)
z_w = z_w / np.sqrt(2 * np.pi)
```

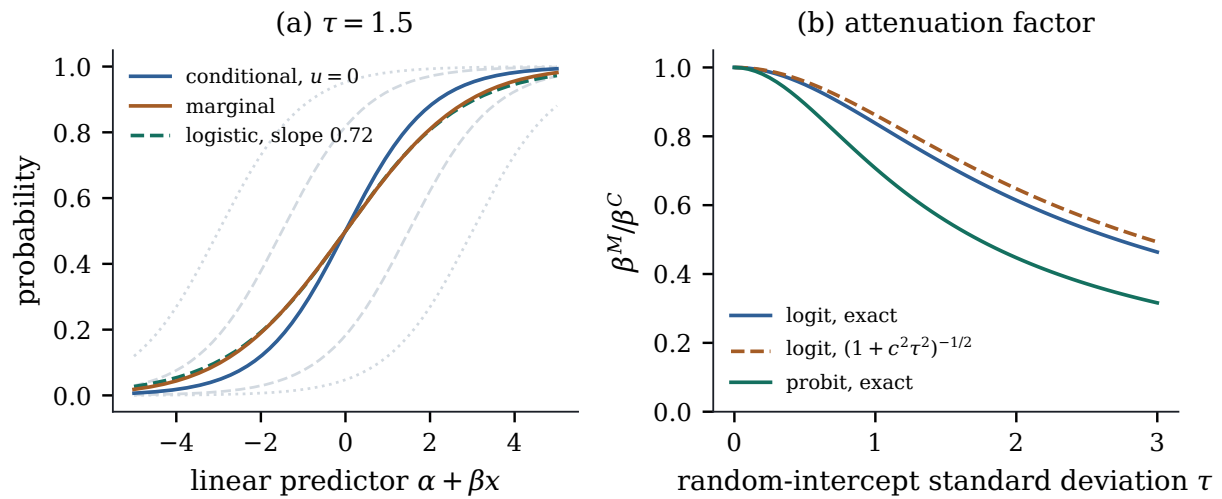


Figure 40.1.1: Attenuation in a random-intercept binary model. (a) With $\tau = 1.5$: five conditional logistic curves (grey), their average, and the logistic curve solving Eq. 40.1.3. (b) The ratio β^M/β^C against τ for the logit (exact and approximate) and for the probit.

```
def marginal_curve(eta, tau, link="logit"):
    """E[H(eta + U)] for U ~ N(0, tau^2), by Gauss-Hermite quadrature."""
    H = expit if link == "logit" else norm.cdf
    grid = np.atleast_1d(eta)[..., None] + tau * z_node
    return np.sum(z_w * H(grid), axis=-1)

def beta_marginal(alpha, beta, tau, link="logit"):
    """Population-averaged coefficients: the root of E[(1, x){m(x) - H(a + b x)}] = 0
    when x ~ N(0, 1), with m the marginal curve of the conditional model."""
    H = expit if link == "logit" else norm.cdf
    m = marginal_curve(alpha + beta * z_node, tau, link)

    def score(par):
        resid = m - H(par[0] + par[1] * z_node)
        return [np.sum(z_w * resid), np.sum(z_w * z_node * resid)]

    sol = root(score, [alpha, beta], tol=1e-13)
    return sol.x

C_LOGIT = 16 * np.sqrt(3) / (15 * np.pi)

def approx_factor(tau, link="logit"):
    c = C_LOGIT if link == "logit" else 1.0
    return 1.0 / np.sqrt(1.0 + c**2 * tau**2)

for t in [0.5, 1.0, 2.0]:
    print(f"tau = {t}: logit {beta_marginal(0.0, 1.0, t)[1]:.4f}"
          f" approximation {approx_factor(t):.4f}"
          f" probit {1 / np.sqrt(1 + t**2):.4f}")
```

[Run this code online](#)

40.1.4 Attenuation is not bias

Attenuation is not bias: both coefficients are correct answers to different questions. The phenomenon is the **non-collapsibility** of the odds ratio, met in [Section 35.5](#) as Example (The odds ratio does not collapse) — averaging 2×2 tables that all share one odds ratio produces a table whose odds ratio is closer to one, with no confounding whatever. Risk differences and risk ratios do not behave this way, which is parts (a) and (b).

It follows that the conditional coefficient depends on what else is in the model: adding a covariate that explains part of the cluster heterogeneity lowers τ^2 and, by [Eq. 40.1.5](#), moves the conditional and marginal coefficients of the *other* covariates closer together. That is not evidence of confounding. Nor is either coefficient a causal quantity by itself: randomization (Theorem 25.2.3) licenses a causal reading of the marginal contrast between treated and untreated, but does not turn the marginal log odds ratio into the conditional one, and does not license a conditional reading of β^C unless the random effect really is a property of the cluster that the intervention leaves alone.

Which target, then? The conditional coefficient answers questions about a change *within a unit* — a dose for a patient, a policy for a school — and is comparable across studies with different amounts of unmeasured heterogeneity, provided the random-effect distribution is right. The marginal coefficient answers questions about a population total or rate, and is safer when the cluster structure is a nuisance, because it assumes nothing about the cluster effects. For the identity link the distinction disappears, which is why [Chapter 32](#) never raised it: in Definition 32.2.1 the marginal model [Eq. 32.2.2](#) has the same $\mathbf{X}\boldsymbol{\beta}$ as the conditional one.

40.1.5 Exercises

40.1.5.1 A. Check your understanding

Exercise. A1

A random-intercept logistic model for whether a pupil passes an examination has $\hat{\beta}^C = 0.60$ for one extra hour of study per week and $\hat{\tau} = 1.2$. Use [Eq. 40.1.6](#) to say roughly what a population-averaged logistic model would report, and state in one sentence each what the two numbers mean.

Solution. The factor is $(1 + 0.5881^2 \times 1.2^2)^{-1/2}$. Now $0.5881^2 = 0.3459$ and $1.2^2 = 1.44$, so the factor is $(1 + 0.4981)^{-1/2} = 1.4981^{-1/2} \approx 0.817$, giving $\hat{\beta}^M \approx 0.49$. The first says: for a given pupil, an extra hour of study multiplies the odds of passing by $e^{0.60} \approx 1.82$. The second says: comparing two randomly chosen pupils from the population who differ by an hour of study, the odds ratio is about $e^{0.49} \approx 1.63$.

Exercise. A2

A Poisson random-intercept model with a log link is fitted and reports $\hat{\alpha} = -1.2$, $\hat{\beta} = 0.35$, $\hat{\tau}^2 = 0.8$. What does a population-averaged log-linear model estimate? Which of the three numbers changes?

40.1.5.2 B. Practice

Exercise. B1

Use [Eq. 40.1.5](#) to show that in a logistic random-intercept model with $\Pr(Y = 1 \mid U) = h(U)$ and U symmetric about 0, the attenuation factor at $\theta = 0$ equals $1 - 4 \operatorname{Var}\{h(U)\}$. Evaluate it for $U \sim N(0, 1)$, for which $\operatorname{Var}\{h(U)\} = 0.0434$, and compare with the ratio 0.8379 of [Example \(Attenuation in a logistic random-intercept model\)](#).

Solution. By symmetry $E h(U) = 1/2$, so the denominator of [Eq. 40.1.5](#) is $1/4$ and $F'(0) = 1 - 4 \operatorname{Var}\{h(U)\}$. With $\operatorname{Var}\{h(U)\} = 0.0434$ this is $1 - 4 \times 0.0434 = 0.8264$. This is the *local* attenuation at $\theta = 0$, and it is the strongest: as $|\theta|$ grows, P_θ concentrates near 0 or 1, its variance vanishes faster than $E P_\theta E(1 - P_\theta)$ and $F' \rightarrow 1$. The global ratio $\beta^M/\beta^C = 0.8379$ of [Example \(Attenuation in a logistic random-intercept model\)](#) is accordingly a little larger, because [Eq. 40.1.3](#) averages F' over the whole range of X .

Exercise. B2

Extend [Proposition 40.1.1\(c\)](#) to a probit model with a random *slope*: $\Pr(Y = 1 \mid U_0, U_1) = \Phi(\alpha + \beta x + U_0 + U_1 x)$ with $(U_0, U_1) \sim N_2(\mathbf{0}, \mathbf{G})$. Show that the marginal curve is $\Phi\{(\alpha + \beta x)/s(x)\}$ with $s(x)^2 = 1 + (1, x)\mathbf{G}(1, x)^\top$, and explain why the marginal model is now not a probit model in x at all.

Solution. Exactly as in the proof of (c), with $Z = U_0 + U_1 x$ normal with mean zero and variance $1 + g_{00} + 2g_{01}x + g_{11}x^2 = s(x)^2$. The marginal linear predictor $(\alpha + \beta x)/s(x)$ is a ratio of a linear function to the square root of a quadratic, which is linear in x only if $g_{11} = g_{01} = 0$. With a random slope the attenuation is different at different x , so no single marginal coefficient describes it.

40.1.5.3 C. Going deeper**Exercise. C1**

[Eq. 40.1.6](#) is an approximation built for a normal random effect. This exercise computes the attenuation exactly for a two-point one and shows how different the answer can be. Take a binary covariate and the model $\logit \Pr(Y = 1 \mid x, U) = \beta x + U$, where $U = \pm c$ with probability $\frac{1}{2}$ each, $c > 0$ and $\beta > 0$.

- (a) Show that $m(0) = \frac{1}{2}$ exactly, so that $\alpha^M = 0$ and β^M is the marginal log odds at $x = 1$.
- (b) Show that

$$\beta^M = \log \left\{ 1 + \frac{e^{2\beta} - 1}{1 + e^\beta \cosh c} \right\} = \beta + \log \frac{\cosh c + e^\beta}{1 + e^\beta \cosh c}.$$

- (c) Check that $\beta^M = \beta$ at $c = 0$, that β^M decreases in c , and that $0 < \beta^M < \beta$ for $c > 0$, as [Proposition 40.1.1\(d\)](#) requires.
- (d) Show that $\beta^M \rightarrow 0$ as $c \rightarrow \infty$, and that $\beta^M \sim 4 \sinh(\beta) e^{-c}$. Since $\operatorname{Var}(U) = c^2$, compare with the prediction of [Eq. 40.1.6](#) at $\tau = c$. What does the comparison say about that formula?

Solution.

- (a) $m(0) = \frac{1}{2}\{h(c) + h(-c)\} = \frac{1}{2}$, because $h(-\eta) = 1 - h(\eta)$ for the logistic h . So $gm(0) = 0$, and by the remark after Eq. 40.1.3 a binary covariate matches the two marginal means exactly, giving $\alpha^M = 0$ and $\beta^M = gm(1)$.
- (b) Write $b = e^\beta$ and $t = e^c$. Then $h(\beta + c) = bt/(1 + bt)$ and $h(\beta - c) = b/(t + b)$, so

$$h(\beta + c) + h(\beta - c) = \frac{b\{t(t + b) + (1 + bt)\}}{(1 + bt)(t + b)} = \frac{b(t^2 + 1 + 2bt)}{t(1 + b^2) + b(1 + t^2)}.$$

Hence $2m(1)$ is that expression, and $2\{1 - m(1)\} = \{2t(1 + b^2) + 2b(1 + t^2) - b(t^2 + 1 + 2bt)\}/\{t(1 + b^2) + b(1 + t^2)\}$, whose numerator simplifies to $2t + b + bt^2$. The marginal odds are therefore

$$\frac{m(1)}{1 - m(1)} = \frac{b(t^2 + 1 + 2bt)}{b(1 + t^2) + 2t} = \frac{b(\cosh c + b)}{1 + b \cosh c},$$

after dividing numerator and denominator by $2t$ and using $(t^2 + 1)/(2t) = \cosh c$. Taking logarithms gives the second form; and $b(\cosh c + b)/(1 + b \cosh c) = 1 + (b^2 - 1)/(1 + b \cosh c)$ gives the first.

- (c) At $c = 0$, $\cosh c = 1$ and the first form is $\log\{1 + (b^2 - 1)/(1 + b)\} = \log b = \beta$. The first form is visibly decreasing in $\cosh c$, hence in c , because $b^2 > 1$; and it stays positive for the same reason. This is Proposition 40.1.1(d) in closed form.
- (d) As $c \rightarrow \infty$, $(b^2 - 1)/(1 + b \cosh c) \rightarrow 0$, so $\beta^M \rightarrow 0$; and $\log(1 + u) \sim u$ with $\cosh c \sim e^c/2$ gives $\beta^M \sim (b^2 - 1)/(b e^c/2) = 2(b - b^{-1})e^{-c} = 4 \sinh(\beta)e^{-c}$. The approximation Eq. 40.1.6, applied at $\tau = c$, gives $\beta/\sqrt{1 + c^2} \cdot 0.5881^2$, which falls only like $1/c$. At $c = 6$, say, the exact attenuation is of order e^{-6} while the formula still reports a third of β . The formula is not a general fact about attenuation: it is a calculation for a normal U , where the tails of the random effect keep some clusters at intermediate probabilities, and it fails for a U that pushes every cluster to the flat ends of the logistic curve. Attenuation depends on the whole distribution of U , not on τ alone.

Exercise. C2

With a random *slope* the marginal coefficient need not even share the sign of the conditional one. Extend Exercise (B2) to the logit: let x be binary and

$$\text{logit } \Pr(Y = 1 \mid x, U_1) = \alpha + \beta x + U_1 x, \quad U_1 = \pm d \text{ with probability } \frac{1}{2},$$

and take $\alpha = \log 3$, $\beta = \log 2$, $d = \log 6$.

- (a) Show that the two clusters have conditional log odds ratios $\beta + d = \log 12$ and $\beta - d = \log \frac{1}{3}$, whose mean is $\beta = \log 2 > 0$.
- (b) Show that $m(0) = \frac{111}{148}$ and $m(1) = \frac{109}{148}$, so that the marginal odds ratio is $\frac{109}{117} < 1$ and $\beta^M < 0$.
- (c) Show that the marginal risk difference is $-\frac{1}{74}$ and equals the average of the two cluster-specific risk differences $\frac{33}{148}$ and $-\frac{37}{148}$, while the marginal log odds ratio is *not* the average of the two conditional ones.

- (d) Explain why [Proposition 40.1.1\(d\)](#) does not apply, and why “attenuation” is the wrong word for what has happened.

Solution.

- (a) Within a cluster, U_1 is fixed, so the log odds at $x = 1$ minus the log odds at $x = 0$ is $\beta + U_1$, namely $\log 2 + \log 6 = \log 12$ or $\log 2 - \log 6 = \log \frac{1}{3}$. The mean over clusters is β because $\mathbb{E} U_1 = 0$.
- (b) At $x = 0$ the random slope drops out, so $m(0) = h(\alpha) = 3/4 = 111/148$. At $x = 1$ the linear predictors are $\alpha + \beta + d = \log 36$ and $\alpha + \beta - d = \log 1 = 0$, so $m(1) = \frac{1}{2}(\frac{36}{37} + \frac{1}{2}) = \frac{1}{2} \cdot \frac{109}{74} = \frac{109}{148}$. The marginal odds are $(109/148)/(39/148) = 109/39$ at $x = 1$ and 3 at $x = 0$, so the marginal odds ratio is $109/117$, less than one.
- (c) $m(1) - m(0) = -2/148 = -1/74$. The cluster-specific risk differences are $\frac{36}{37} - \frac{3}{4} = \frac{144-111}{148} = \frac{33}{148}$ and $\frac{1}{2} - \frac{3}{4} = -\frac{37}{148}$, whose average is $(33 - 37)/296 = -1/74$. Equality holds because $m(1) = \mathbb{E} h(\alpha + \beta + U_1)$ is itself an average, so risk differences pass through it; log odds ratios do not, and $\log(109/117) \neq \frac{1}{2}\{\log 12 + \log \frac{1}{3}\} = \log 2$.
- (d) Part (d) of the proposition concerns $h(\theta + U)$, a random *level* added to a common linear predictor, and its proof differentiates $\theta \mapsto g \mathbb{E} h(\theta + U)$. Here the random term multiplies x , so the two conditional curves have different slopes and the argument does not start. The conclusion fails too: β^M and β^C have opposite signs. Calling the difference attenuation would suggest that the marginal coefficient is a shrunken version of the conditional one, and here it is not a version of it at all — β^C is an average on the log odds scale, dominated by the cluster in which the treatment helps a great deal, while $m(1)$ is an average of probabilities, in which the cluster where the treatment *hurts* costs more than the other gains. Two averages on two scales, of a quantity that varies across clusters, and they need not agree even in sign; the warning after [Eq. 40.1.3](#) applies with full force.